

Ahmad Aldirieh

+1 (226) 505-4534 | a2aldiri@uwaterloo.ca | [ahmadaldirieh.com](mailto:a2aldiri@uwaterloo.ca) | [linkedin.com/in/ahmad-dirieh](https://www.linkedin.com/in/ahmad-dirieh)

SKILLS

Technical Skills: EPLAN ProPanel, Arduino, 3D Modelling, PCB Design (TraxMaker), Circuit Board Assembly, Soldering, Machine Learning, Siemens TIA Portal 20, Altium Designer, Excel, Verilog, Git

Programming Languages: C++, Python, HTML, CSS, JavaScript, Processing, Great Cow BASIC, Ladder Logic

Soft Skills: Leadership, Problem Solving, Critical Thinking, Project Coordination, Multitasking, Attention to Detail

Languages: English (Fluent), Arabic (Fluent)

WORK EXPERIENCE

Control Systems Hardware Design Co-op

Jan 2026 - May 2026

ATS Corporation-Industrial Automation

- Designed and refined electrical schematics in EPLAN, implementing design changes to ensure system functionality and alignment with project requirements.
- Supported development of detailed Bills of Materials (BOM), conducting component research, cross-referencing vendors, and resolving hardware constraints to maintain system performance.
- Collaborated cross-functionally with software engineers and shop-floor electricians to troubleshoot issues, drive design updates, and meet tight project deadlines, including extended project support.

Electrical Engineering Design Co-op

Jun 2024 - Jul 2024

Rittal Limited Canada

- Designed electrical motor control centers (Ri4Power) using EPLAN, developed 3D macros, designed copper busbar systems, managed part numbers and documentation to ensure project accuracy.
- Supported daily operations at the Modification Centre by coordinating job assignments, verifying parts, and assisting with the assembly of the enclosures and order preparation.
- Participated in meetings with clients and suggested solutions to resolve technical issues and workflow improvements that enhanced accuracy and efficiency.

University of Waterloo Formula Electric - Electrical Subsystem Member

Sept 2025 - Present

University of Waterloo

- Designed PCB boards using Altium Designer, contributing to reliable electrical systems for the team's race car.
- Collaborated on subsystem integration, troubleshooting, and hands-on implementation in a competitive environment.
- Strengthened problem-solving and project delivery skills through technical teamwork.

Student Researcher

Nov 2024 - Feb 2025

University of Waterloo

- Assisted Dr. Eugene Li's lab on predictive maintenance for CNC mill blades by preparing datasets and supporting a machine learning model for RUL estimation.
- Tested outputs and documented results to ensure model reliability and practical application.
- Contributed insights that supported maintenance planning and reduced CNC downtime.

PROJECTS

Inferno-1400 (Firefighter Robot)

Sept 2024 - Jan 2025

Laurel Heights Secondary School

- Co-developed project proposal with timeline, materials, and 3D AutoCAD designs.
- Built autonomous robot with line, flame, and wall detection sensors to meet competition requirements.
- Programmed robot to navigate mazes and extinguish flames, demonstrating coding, problem-solving, and collaboration.

EDUCATION

University of Waterloo

Sept 2025 - Apr 2030

Bachelor of Applied Science in Electrical Engineering

Related Courses: Digital Circuits and Systems (ECE 124), Electricity and Magnetism (ECE 106)

CERTIFICATIONS & AWARDS

Certifications: EPLAN Essential Training & EPLAN Advanced Applications Training

Awards: University of Waterloo's President's Scholarship, Faculty of Engineering Entrance Scholarship (2025 & 2026)